

Milad Ansari

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Education

Shahid Beheshti University

Bachelor of Computer Engineering

Cumulative GPA: 16.87/20 (US grading system 3.37/4)

Sep 2022 – Present

Alborz High school

Diploma in Mathematics-Physics

Cumulative GPA: 19.05/20 (US grading system 4/4)

Sep 2019 – Jun 2022

Research Interests

Internet of Things (IoT)
Embedded Systems

Edge Computing
Embedded AI

Honors and Awards

ICPC Asia West Regional Contest : Bronze Medal **Certificate**

Spring 2024

ICPC Asia West Regional Contest : Bronze Medal **Certificate**

Fall 2023

Amirkabir Collegiate Programming Contest(ACPC) : First Place

Spring 2022

Iranian National Informatics Olympiad : Silver Medal

Summer 2021

– Issued by **Young Scholars Club** & **NODET**

Projects

smart-home-knx-thingsboard **GitHub**

- Smart Home IoT System (KNX, ThingsBoard, AI) – Simulated a KNX home in KNX Virtual/ETS6 and bridged it to ThingsBoard via a customized IoT Gateway KNX connector, with web/mobile telemetry and bidirectional RPC control.
- Local AI assistant (MCP + Ollama) for natural-language monitoring and control.

esp32s3-thingsboard-mqtt-onboarding **GitHub**

- Secure IoT Device Onboarding (ESP32-S3, ThingsBoard) – Built C++ firmware for secure MQTT-over-TLS telemetry with automated device provisioning and claiming, backed by a self-hosted ThingsBoard CE server (Docker, PostgreSQL); added an optional LVGL UI for on-device status display.

embedded-heating-system-simulink **GitHub**

- 4-Room Heating System (MATLAB/Simulink) – Deployed a model-based design (MBD) workflow to simulate and control thermal dynamics. Designed a 6-state Stateflow controller with a safety checker pattern, generated production-ready C code via Embedded Coder, and validated logic on an Arduino board using the Wokwi simulator.

MPU6050-Edge-AI-Motion-Classification **GitHub**

- AIoT motion classification system using simulated MPU6050 sensor data, Edge Impulse WebAssembly inference, MQTT, and ThingsBoard for real-time monitoring and alerts.

Experience

ICPC Student Team, Shahid Beheshti University

Oct 2023 – Present

- Member of the university's ICPC competitive programming team, representing SBU in regional contests.

Shahid Beheshti University(Volunteer Teaching Assistant)

Feb 2023 – Present

Microprocessors and Integrated Circuits – **M. Dara Rahmati**

- Spring 2026

Microprocessors and Integrated Circuits – **M. Mohammad Hosseini**

- Spring 2026

Computer Architecture – **M. Dara Rahmati**

- Fall 2025(Head TA) – Fall 2024

Algorithm Design – **Ms. Ramak Ghavamizadeh Meibodi**

– Spring 2025(Head TA)

Discrete Mathematics – **Ms. Monireh Abdoos**

– Fall 2024

Data Structures – **M. Mehran Alidoost Nia**

– Spring 2024

Advanced Programming – **M. Mojtaba Vahidi-Asl**

– Spring 2023

Work Experience(Instructing students in IOI-algorithmic problem-solving)

Feb 2021 – May 2025

Allameh Helli "NODET" High School

– Spring 2025

– Fall 2024

– Spring 2024

– Fall 2023

– Summer 2022

Amirkabir University Of Technology(Algorithm Arena Course)

– Summer 2024

Alborz High School

– Summer 2024

– Summer 2023

– Fall 2021

Related Course

Internet of Things : 20

Fundamentals of Real-Time and Embedded Systems : 18

Microprocessors and Integrated Circuits : 20

Computer Architecture : 19.52

Data Structures : 20

Design of Algorithms : 19.9

Skills

Languages: C/C++, Python, Java, Assembly (MIPS, 8086, ARM), MySQL, Verilog

Embedded: ESP32, Arduino, STM32

Others: MATLAB, Simulink, Git, Linux, Docker